

1. What are the basic components of an Agent? (2)
2. Prove or disprove the proposition P using a typical forward chaining algorithm on the knowledgebase that follows. (3)
 $A, B, C, D, A \wedge G \Rightarrow H, A \wedge H \Rightarrow E, C \wedge K \Rightarrow L, H \wedge K \Rightarrow P, B \wedge E \Rightarrow G, B \wedge D \Rightarrow K,$
 $D \wedge F \Rightarrow G, D \wedge H \Rightarrow F, D \wedge L \Rightarrow H.$
3. How a Term is defined in the syntax of First Order Logic? (2)
4. Illustrate with an example the major steps of a typical unification algorithm. (3)

1. When an Agent is said to be rational? (2)
2. Prove or disprove the proposition H using a typical backward chaining algorithm on the knowledgebase that follows. (3)
 $A, B, C, D, A \wedge G \Rightarrow H, A \wedge H \Rightarrow E, C \wedge K \Rightarrow L, H \wedge K \Rightarrow P, B \wedge E \Rightarrow G, B \wedge D \Rightarrow K,$
 $D \wedge F \Rightarrow G, D \wedge H \Rightarrow F, D \wedge L \Rightarrow H.$
3. How do you define the syntax and semantics of First Order Logic? (2)
4. Describe the major steps of proving a natural language sentence using resolution on a knowledgebase. (3)

1. What are the main issues in specification of task environment for designing a rational agent? (2)
2. Illustrate with an example the fact that forward chaining algorithms are data driven. (3)
3. Explain the matching rules that are involved during Unification. (2)
4. Convert the following FOL sentence into CNF. (3)
 $\forall x (\text{Student}(x) \wedge \text{Takes}(x, \text{CSE4107}) \leftrightarrow \text{CSE_Student}(x)).$

1. What are the basic types of Agents? (2)
2. Explain with an example that backward chaining algorithms are said to be goal oriented. (3)
3. Show that $\exists x (\text{Student}(x) \wedge \text{Plays}(x, \text{Cricket}))$ is a valid sentence of First Order Logic. (2)
4. Derive $P2(\text{Best}, \text{Rest})$ from the following clauses using the Resolution rule systematically. (3)
 $P1(x, F1(y)) \vee P2(x, u), \neg P1(\text{Best}, z) \vee P3(u, F2(x)), \neg P3(\text{Rest}, v).$

1. How do you explain the concept of Percepts and Actions of an Agent? (2)
2. Explain the characteristics of forward chaining algorithms. (3)
3. What are the major steps of conversion of FOL sentences to CNF? (2)
4. Derive a most general unifier for the following set of sentences. (3)
 $S = \{P2(x, F1(\text{Mia}), \text{Shaheb}, y), P2(\text{Babu}, v, u, F2(x))\}.$

1. What contrasting features of environment are generally considered for designing rational agents? (2)
2. Explain the characteristics of backward chaining algorithms. (3)
3. How do you explain the concept of unification? (2)
4. Show that
 $\forall x, y, z ((\text{Brother}(x, y) \wedge \text{Son}(z, x)) \Rightarrow \text{Nephew}(z, y))$
is a syntactically correct sentence of First Order Logic. (3)

1. Write a brief illustrative essay on Actions and Actuators of an Agent. (3)
2. Prove or disprove the proposition S using a typical backward chaining algorithm on the knowledgebase that follows. (4)

$$A, B, C, D, A \wedge G \Rightarrow P, A \wedge P \Rightarrow E, C \wedge Q \Rightarrow M, P \wedge R \Rightarrow S, B \wedge E \Rightarrow G, B \wedge D \Rightarrow Q, \\ D \wedge F \Rightarrow G, D \wedge P \Rightarrow F, D \wedge M \Rightarrow P, A \wedge B \Rightarrow R.$$
3. How do you explain the concept of Unification? (3)

1. Write a brief illustrative essay on Sensors and Percepts of an Agent. (3)
2. Demonstrate that

$$\forall x, y, z ((\text{Father}(z, x) \wedge \text{Father}(z, y) \wedge \text{Male}(x) \wedge \neg(x = y)) \Rightarrow \text{brother}(x, y))$$
is a syntactically correct FOL sentence. (4)
3. How Resolution is used to prove or disprove a sentence? (3)

1. Discuss the Environment issues that are considered for designing rational agents. (3)
2. Convert the following English sentence to CNF of FOL. (4).
A student of CSE department must take CSE1101 and CSE4107.
3. Distinguish between forward and backward chaining algorithms. (3)

1. Elaborate the facts that make an Agent rational. (3)
2. Prove or disprove $\exists x(\text{Math}(x) \wedge \text{Takes}(\text{Karim}, x))$ from the following knowledgebase using resolution. (4)
 - a) $\text{Student}(\text{Karim})$
 - b) $\text{Math}(\text{MathFor}(y)) \vee \neg \text{Student}(y)$
 - c) $\neg \text{Student}(y) \vee \text{Takes}(y, \text{MathFor}(y))$
3. Describe a situation when inferences based on pure logic fails in choosing from alternatives. (3)

1. How do you characterize the knowledgebase for reasoning with propositional logic? (3)
2. What is an mgu (most general unifier)? Derive an mgu, if possible, for the following set of FOL sentences. (1+3)

$$S = \{P1(F2(\text{Miki}), z, \text{Blue}, u), P1(y, \text{Red}, x, F3(z))\}.$$
3. Explain with an illustrative example the semantics of FOL. (3)

1. Explain the characteristics of forward chaining algorithms. (3)
2. Take the following situation of 'Find Gold avoiding Monsters' problem, and derive $\neg M_{3,2}$. (4)

$$S_{i,j} \Leftrightarrow M_{i-1,j} \vee M_{i+1,j} \vee M_{i,j-1} \vee M_{i,j+1} \quad (i \geq 1) \wedge (j \geq 1) \wedge (i \leq 4) \wedge (j \leq 4) \\ \neg M_{3,1} \wedge \neg S_{3,1} \wedge \neg G_{3,1} \quad A_{3,1}$$
3. How Atomic Sentences are defined in the syntax of FOL? (3)

1. Discuss about possible 'Sensors' and 'Actuators' of an automatic car. (3)
2. Prove or disprove the proposition P using a typical forward chaining algorithm on the knowledgebase that follows. (4)

$$A, B, C, D, G \wedge P \Rightarrow H, A \wedge H \Rightarrow E, C \wedge K \Rightarrow L, H \wedge K \Rightarrow S, B \wedge F \Rightarrow G, B \wedge D \Rightarrow K, \\ D \wedge F \Rightarrow G, D \wedge G \Rightarrow P, D \wedge L \Rightarrow H.$$
3. Explain resolution-refutation completeness of a knowledgebase. (3)

1. How do you explain that rational Agents do the 'right' thing? (3)
2. Demonstrate that

$$\forall x, y, z ((\text{Mother}(z, x) \wedge \text{Mother}(z, y) \wedge \text{Female}(x) \wedge \neg(x = y)) \Rightarrow \text{Sister}(x, y))$$
is a syntactically correct FOL sentence. (4)
3. Write a brief essay on Unification. (3)

1. What are the basic types of agents? (2)
2. Why backward chaining algorithms are said to be goal oriented? (2)
3. Explain the meaning of semantics of FOL. (2)
4. Prove or disprove $\exists x(\text{Course}(x) \wedge \text{Takes}(\text{Hasan}, x))$ from the following knowledgebase using resolution. (4)
 a) $\text{Student}(\text{Hasan})$ b) $\text{Course}(\text{CourseFor}(x)) \vee \neg \text{Student}(x)$ c) $\neg \text{Student}(x) \vee \text{Takes}(x, \text{CourseFor}(x))$

1. Explain the concept of Percepts of an Agent. (2)
2. Illustrate the fact that an agent making inferences based on pure logic may fail in choosing from alternatives even in a simple situation. (3)
3. What are the matching criteria for Unification? (2)
4. Convert the following English sentence to CNF of FOL. (3)
 Every student must take at least one course of HUM type.

1. Why learning capability is a must for a rational Agent? (2)
2. Illustrate with an example how resolution is used to prove or disprove sentences? (3)
3. Demonstrate that

$$\forall x, y, z ((\text{Brother}(x, y) \wedge \text{Son}(z, x)) \Rightarrow \text{Nephew}(z, y))$$
is a syntactically correct FOL sentence. (3)
4. How do you characterize forward chaining algorithms? (2)

1. Explain the differences between forward and backward chaining algorithms. (3)
2. Consider the following situation of 'Find Gold avoiding Monsters' problem.

$$S_{i,j} \Leftrightarrow M_{i-1,j} \vee M_{i+1,j} \vee M_{i,j-1} \vee M_{i,j+1} \quad (i \geq 1) \wedge (j \geq 1) \wedge (i \leq 4) \wedge (j \leq 4)$$

$$\neg M_{1,1} \wedge \neg M_{2,1} \wedge S_{2,1} \wedge \neg G_{2,1} \quad A_{2,1}$$
Derive the fact that $M_{3,1} \vee M_{2,2}$ using inference rules of propositional logic. (4)
3. Describe the major steps of conversion of FOL sentences to CNF. (3)